

Bayesian Hypothesis test, Bayes factor and Information criteria

STA 427/527, Fall 2019, Xin Wang

5.1 Tests in Bayesian framework

5.1.1 Frequentist tests

1. Hypothesis:

$$H_0 : \mu = \mu_0 \quad H_A : \mu \neq \mu_0$$

2. Test statistics

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \quad t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \quad F = \frac{MSR}{MSE}$$

Regression, ANOVA, F-stat

3. Sampling distribution under H_0

4. Critical region or p-value → Based on observed Test Statistic & the sampling distribution, we can get p-value & make a decision

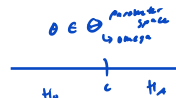
5.1.2 The Bayesian Approach (one-sided case)

- We just need the posterior probability that the parameter θ falls in H_0 and H_A ⇒ we can calculate the actual probability based on the posterior sample

- Hypothesis

→ split parameter space into the null space & alternative space (i.e. subspaces), these could be bounded as well

$$H_0 : \theta \leq c \quad H_A : \theta > c$$



- $\Theta_0 = (-\infty, c]$ and $\Theta_A = (c, +\infty)$.

- Under posterior distribution:

→ can be density or mass function
→ The posterior distribution is $p(\theta|y) \propto p(y|\theta) \pi(\theta)$

$$\rightarrow p(\theta \in \Theta_0 | y) = \int_{-\infty}^c p(\theta | y) d\theta = P_0$$

$$\rightarrow p(\theta \in \Theta_A | y) = \int_c^{\infty} p(\theta | y) d\theta = P_1 = 1 - P_0$$

⇒ Then compare P_0 & P_1 & we select in favor of the larger one

- **Example 1:** Let Y_i be the number of strikes in a sequence of strikes before the cessation of the series and $Y_i | \theta \sim \text{geometric}(\theta)$, for $i = 1, \dots, 11$. The pmf is

$$p(y) = \theta(1 - \theta)^y, \quad y = 0, 1, 2, \dots$$

And it is known that $\sum_{i=1}^{11} y_i = 148$. ↳ # of failures parameterization

1. The prior distribution is $p(\theta) \propto \sqrt{I(\theta)}$

Jeffrey's Prior, we know the conjugate prior for geometric is Beta, but we want a non-informative prior for this example

Finding Jeffrey's prior

$$\rightarrow L(\theta | y) = \prod_{i=1}^n \theta(1 - \theta)^{y_i} = \theta^n (1 - \theta)^{\sum_{i=1}^n y_i}, \quad n = 11$$

$$\rightarrow \ell(\theta | y) = n \log(\theta) + \sum_{i=1}^n y_i \log(1 - \theta)$$

* Make sure to use the joint dist of the data, not on individual y_i

$$\rightarrow \frac{d \ell(\theta|Y)}{d\theta} = \frac{n}{\theta} - \frac{\sum y_i}{1-\theta}$$

$$\rightarrow \frac{d^2 \ell(\theta|Y)}{d\theta^2} = \frac{-n}{\theta^2} - \frac{\sum y_i}{(1-\theta)^2}$$

$$\rightarrow \mathcal{I}(\theta) = -E_Y \left[\frac{d^2 \ell(\theta|Y)}{d\theta^2} \right] = \frac{n}{\theta^2} + \frac{n \left(\frac{1-\theta}{\theta} \right)}{(1-\theta)^2}$$

(constant w/ respect to Y)

$$= \frac{n}{\theta^2} + \frac{n}{\theta(1-\theta)}$$

$$= \frac{n}{\theta^2(1-\theta)}$$

$$\leftarrow \text{write } E(y_i) = \frac{1-\theta}{\theta}$$

$$\rightarrow \pi(\theta) \propto \sqrt{\mathcal{I}(\theta)}$$

$$= \sqrt{\frac{n}{\theta^2(1-\theta)}}$$

$$\propto \theta^{-1} (1-\theta)^{-1/2}$$

↳ cannot write as a Beta dist b/c of the θ^{-1} (i.e. can't specify Beta(0, ...))

2. Posterior of θ

$$\rightarrow \text{Data distribution} \rightarrow P(Y| \theta) = \prod_{i=1}^n \theta(1-\theta)^{y_i} = \theta^n (1-\theta)^{\sum y_i}$$

$$\rightarrow \text{prior dist} \rightarrow \pi(\theta) \propto \theta^{-1} (1-\theta)^{-1/2}$$

$$\Rightarrow \text{posterior dist} \rightarrow P(\theta|Y) \propto P(Y|\theta) \pi(\theta)$$

$$= \theta^n (1-\theta)^{\sum y_i} \cdot \theta^{-1} (1-\theta)^{-1/2}$$

$$= \theta^{n-1} (1-\theta)^{\sum y_i - 1/2}$$

$$= \theta^{n-1} (1-\theta)^{(\sum y_i + 1/2) - 1}$$

$$\sim \text{Beta}(n, \sum y_i + 1/2) \quad n=11, \sum y_i = 148$$

$$\Rightarrow \sim \text{Beta}(11, 148.5)$$

$$3. H_0: \theta \leq 0.05, \quad H_A: \theta > 0.05$$

$$\rightarrow H_0: P(\theta \leq 0.05 | Y) = \text{pbeta}(x=0.05, \text{shape1}=11, \text{shape2}=148.5) = 0.1715$$

$$\rightarrow H_A: P(\theta > 0.05 | Y) = 1 - 0.1715 = 0.8285$$

$\Rightarrow$ In favor of H_A

4. Simulation

$$\rightarrow \text{ML approach} \rightarrow \text{in R: } \text{post_theta} = \text{rbeta}(n=10000, \text{shape1}=11, \text{shape2}=148.5)$$

$$\Rightarrow H_0: P(\theta \leq 0.05) = \text{mean}(\text{post_theta} \leq 0.05)$$

$$5. H_0: \theta = 0.1 \quad H_A: \theta \neq 0.1$$

$\rightarrow$ Two-tailed test

Another example

$\rightarrow$ In the regression problem: $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon$

$\leftarrow \dots \rightarrow$

$\rightarrow$ Posterior samples for β_1 & β_2

$$\begin{matrix} \beta_1^{(1)} & \beta_2^{(1)} \\ \vdots & \vdots \\ \beta_1^{(m)} & \beta_2^{(m)} \end{matrix}$$

$\rightarrow \alpha = 0.05$, construct 95% HPD credible set for β_1 & β_2 separately (in R: `credible`)

$$\Rightarrow [L_{\beta_1}, U_{\beta_1}] \quad + \quad [L_{\beta_2}, U_{\beta_2}]$$

$\Rightarrow$ Check if 0 is in the sets? If not, the parameter is "significant"

$\rightarrow$ For continuous random variables, $P(\theta = c | Y) = 0$
↳ a constant

$\Rightarrow$ So instead, define a $100(1-\alpha)\%$ HPD credible set

+ Reject " $\theta = c$ " if level α if c falls outside of this set

$\rightarrow$ in R: `hpd(gbeta, shape1=11, shape2=148.5) = [0.0327, 0.100]`

$\Rightarrow$ cannot reject H_0

5.2 Bayes Factor

- Bayes Factor is used to compare models (+ also for some tests), can compare any models (its a general method)
(i.e. can be totally different models, unlike in ...)
- Compare two models M_1 and M_2 . Given data y , consider the two models, M_1 and M_2 are not necessarily nested.

$$M_1 : p_1(y|\theta_1, M_1)$$
$$M_2 : p_2(y|\theta_2, M_2)$$

→ In frequentist, we can use the LRT, but we must have nested models, where model H_0 is a special case of the alternative model

- Priors: $p(\theta_1|M_1), p(M_1), p(\theta_2|M_2), p(M_2)$ → often, we don't have any preference for what the prior tells us → so we just set it equal for both models
- Bayes factor is defined as

↳ Most of the time, the Bayes Factor (BF) is not easy to calculate b/c finding the distribution $f(y)$ over our dataset is usually very hard. There are packages w/ algorithms to help.

$$BF_{12} = \frac{p(y|M_1)}{p(y|M_2)} \rightarrow \left. \begin{array}{l} \text{Models 1 + 2} \\ \text{Data} \end{array} \right\} \text{the num + denom are the marginal dists of } y \text{ under the two models}$$

⇒ Essentially, BF compares two marginal distn + determines from which model the data is more likely to come from

→ BF_{12} = ratio of marginal dists of y according to each model

(end result → this is how to derive this)

(when integrate θ_i out of the joint dist, it's just the marginal of y)

$$= \frac{\int p(y, \theta_1 | M_1) d\theta_1}{\int p(y, \theta_2 | M_2) d\theta_2}$$

(rewrite as conditional * prior)

$$= \frac{\int p(y|\theta_1, M_1) p(\theta_1|M_1) d\theta_1}{\int p(y|\theta_2, M_2) p(\theta_2|M_2) d\theta_2}$$

data distributions priors of θ_1 & θ_2

$$= \frac{\int (\text{posterior of } \theta_1) d\theta_1}{\int (\text{posterior of } \theta_2) d\theta_2}$$

(In Bayes, generally we have a data dist + a prior)

$$\Rightarrow \text{posterior } p(\theta|y) = \frac{\text{joint dist of } y + \theta}{\text{marginal of } y} = \frac{p(y, \theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{p(y)}$$

⇒ so w/ $\frac{p(y|\theta_1)}{p(y|\theta_2)}$, the $p(y)$ cancels out when re-writing + we are just left w/ the numerators

→ Relationship to posterior probabilities of the two models

$$\rightarrow \frac{p(M_1|y)}{p(M_2|y)} = \frac{p(M_1, y)/p(y)}{p(M_2, y)/p(y)} = \frac{p(M_1, y)}{p(M_2, y)}$$



$$= \frac{p(y|M_1)p(M_1)}{p(y|M_2)p(M_2)}$$
$$= BF_{12} \times \frac{p(M_1)}{p(M_2)}$$

$$\Rightarrow BF_{12} = \frac{p(M_1, y)}{p(M_2, y)} \bigg/ \frac{p(M_1)}{p(M_2)}$$

↓
the density functions
↓
= kinda like a weight for the preference of priors

- If $p(M_1) = p(M_2)$

$$\Rightarrow BF_{12} = \frac{p(M_1, y)}{p(M_2, y)}$$

- Rule of Thumb for BF

→ Suggestions from Jeffreys' opinion in literature (in Jeffreys' own words)

BF	Conclusion
$BF_{0A} \geq 1$	M_1 supported
$10^{-1/2} \leq BF_{0A} < 1$	Minimal evidence against M_1
$0.1 \leq BF_{0A} < 10^{-1/2}$	Substantial evidence against M_1
$0.01 \leq BF_{0A} < 0.1$	Strong evidence against M_1
$BF_{0A} < 0.01$	decisive evidence against M_1

≈ 0.916

< notation: $BF_{0A} = BF_{12}$ >

→ Simple tests → $H_0: P(A=B)=1$
 → In this context → $H_0: P(Y|A)=P_0$
 $H_A: P(Y|A) \neq P_0$ } prior to these Bayes factor notes, we did not know how to solve this kind of probability

• Example 1: $Y_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$, $i=1, \dots, n$

– σ^2 is known, we want to test $H_0: \mu = \mu_0$ $H_A: \mu = \mu_A$. Define $BF_{0A} \rightarrow \frac{P(Y|A_0)}{P(Y|A_A)}$

→ we don't need any integrals to solve this b/c it's a simple test (i.e. only checking on values)

⇒ just use a ratio of joint density functions, equivalent to LRT (always)

→ Data dist under H_0

$$P(Y|A=P_0) = \prod_{i=1}^n P(y_i|A_0) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - \mu_0)^2\right\}$$

→ Data dist under H_A

$$P(Y|A=P_A) = \prod_{i=1}^n P(y_i|A_A) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - \mu_A)^2\right\}$$

$$\Rightarrow BF_{0A} = \frac{\exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu_0)^2\right\}}{\exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu_A)^2\right\}} = \exp\left\{-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_0)^2 - \sum_{i=1}^n (y_i - \mu_A)^2\right)\right\}$$

– If σ^2 is known, but we want to test $H_0: \mu = \mu_0$ $H_A: \mu \neq \mu_0$

→ This is not easy to test b/c it's not a simple test (it's a composite test)

⇒ we have to integrate out the parameters, which can be hard

In practice, we still always use this method of taking integrals when comparing models b/c there are some R packages that can help w/ it

$$\rightarrow \text{under } H_0: P(Y|H_0) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu_0)^2\right\}$$

$$\rightarrow \text{under } H_A: P(Y|H_A) = \int_{\mu \in H_A} P(Y|\mu) \pi(\mu) d\mu, \text{ where } P(Y|\mu) = \prod_{i=1}^n P(y_i|\mu)$$

$$\Rightarrow BF_{0A} = \frac{P(Y|H_0)}{P(Y|H_A)}$$

→ don't need to worry about ratio of priors b/c under H_0 , $P(A=\mu_0)=1$ (we are assuming it's exactly this value) & under H_A , $P(A=\mu_A)=1$ b/c it's a continuous RV

• Example 2: 2003: HIV vaccine vs placebo given to 16395 individuals. 2006: HIV test

– 74 HIV positive reported in 8198 individuals receiving placebo

– 51 HIV positive reported in 8197 individuals receiving treatment

– Let p_1 and p_2 denote the probability of HIV positive in placebo and treatment population.

$$H_0: p_1 = p_2 \quad H_A: p_1 \neq p_2$$

– In frequentist framework

→ sample size is large, so we can use normal approximation

$$\rightarrow Z^* = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{\frac{74}{8198} - \frac{51}{8197}}{0.01359} = 2.639$$

$$\begin{aligned} n_1 &= 8198 & n_2 &= 8197 \\ \hat{p}_1 &= \frac{74}{8198} & \hat{p}_2 &= \frac{51}{8197} \\ \hat{p} &= \frac{74+51}{8198+8197} \end{aligned}$$

⇒ at $\alpha=0.05$, reject H_0

– Bayes Factor

$$\begin{aligned} \rightarrow \text{under } H_0: Z^* | p_1, p_2 &\sim N(0,1) \quad \text{or } N\left(\frac{p_1 - p_2}{0.01359}, 1\right) \quad \text{if } p_1 = p_2 \\ \rightarrow \text{under } H_A: Z^* | p_1, p_2 &\approx N\left(\frac{p_1 - p_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, 1\right) \\ &\downarrow \\ &= N\left(\frac{p_1 - p_2}{0.01359}, 1\right) \end{aligned}$$

The idea → instead of deriving the marginal dist of the data, we calculate the marginal dist of the test statistic Z^*

→ we don't need to integrate μ_1 & μ_2 separately, we can just define a new parameter $\theta = \mu_1 - \mu_2$

⇒ we are interested in the difference, not the specific values of each

$$\rightarrow \theta = \frac{\mu_1 - \mu_2}{0.01354} \quad \begin{cases} H_0: \mathbb{Z}^* / \theta \stackrel{\text{approx}}{\sim} \mathcal{N}(0, 1) & [\theta = 0] \\ H_A: \mathbb{Z}^* / \theta \stackrel{\text{approx}}{\sim} \mathcal{N}(0, 1) & [\theta \neq 0] \end{cases}$$

$$\rightarrow BF_{0A} = \frac{P(\mathbb{Z}^* | \mu_1)}{P(\mathbb{Z}^* | \mu_2)} \rightarrow P(\mathbb{Z}^* | \mu_1) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{\mathbb{Z}^{*2}}{2}\right\} = \text{dnorm}(\mathbb{Z}^*)$$

$$\rightarrow P(\mathbb{Z}^* | \mu_2) = \int_{\theta \neq 0} P(\mathbb{Z}^* | \theta) \pi(\theta) d\theta = \int \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(\mathbb{Z}^* - \theta)^2\right\} \pi(\theta) d\theta \quad \text{c depends on the prior we choose obviously?}$$

→ Suppose $\mu_1, \mu_2 \in [\mu_1, \mu_2] \in [4, 6.2]$, let's say we choose $\mu_1 - \mu_2 \in [0, 0.02]$

⇒ this is the difference we are interested in (just picked an upper bound)

⇒ need the chosen interval in terms of θ

$$0 \leq \mu_1 - \mu_2 \leq 0.04 \quad , \quad \text{if } \theta = \frac{\mu_1 - \mu_2}{0.01354} \Rightarrow \mu_1 - \mu_2 = 0.01354 \theta \rightarrow$$

$$0 \leq 0.01354 \theta \leq 0.04$$

$$0 \leq \theta \leq 2.94 \quad , \quad \text{here is our new interval of interest}$$

→ select uniform prior for simplicity $\Rightarrow \pi(\theta) = \frac{1}{2.94} \quad , \quad \theta \in [0, 2.94]$

$$\rightarrow P(\mathbb{Z}^* | \mu_2) = \int_0^{2.94} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(\mathbb{Z}^* - \theta)^2\right\} \frac{1}{2.94} d\theta \quad \text{(This is just a CDF, } \int_0^x f(t) dt = F(x) - F(0) = P(X \leq x))$$

$$\downarrow$$

$$= \frac{1}{2.94} \left[\int_{-\infty}^{2.94} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(\mathbb{Z}^* - \theta)^2\right\} d\theta - \int_{-\infty}^0 \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(\mathbb{Z}^* - \theta)^2\right\} d\theta \right] \quad \text{c } \int_a^b f(x) dx = F(b) - F(a)$$

$$= \frac{1}{2.94} \left[\text{pnorm}(2.94, \text{mean} = \mathbb{Z}^*, \text{sd} = 1) - \text{pnorm}(0, \mathbb{Z}^*, 1) \right]$$

$$\Rightarrow BF_{0A} = \frac{\text{dnorm}(\mathbb{Z}^*)}{\text{pnorm}(2.94, \mathbb{Z}^*, 1) - \text{pnorm}(0, \mathbb{Z}^*, 1)} \quad \text{(just plug in } \mathbb{Z}^* \text{ that was already calculated based on the data given)}$$

• Some extensions

– From Rouder et al (2009), Bayesian t tests for accepting and rejecting the null hypothesis, Psychonomic Bulletin & Review, 16(2), 225–237.

– One sample t-test: $Y_i \stackrel{iid}{\sim} N(\mu, \sigma^2), i = 1, \dots, n$

* $H_0: \mu = \mu_0$ and $H_A: \mu \neq \mu_0$

* Defined $\delta = \frac{\mu - \mu_0}{\sigma}$, where the prior of δ is Cauchy distribution.

* And $\pi(\mu, \sigma^2) \propto (\sigma^2)^{-1}$

* Consider $t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}$

* The reference gave a form for the corresponding BF_{0A} , which is a function of t .

– Two sample t-test: $X_i \stackrel{iid}{\sim} N(\mu_1, \sigma^2), i = 1, \dots, n_1$ and $Y_j \stackrel{iid}{\sim} N(\mu_2, \sigma^2), j = 1, \dots, n_2$

* $H_0: \mu_1 = \mu_2, \quad H_A: \mu_1 \neq \mu_2$

* Priors: A Cauchy prior on $\delta = \frac{\mu_1 - \mu_2}{\sigma}$

* $\pi(\mu, \sigma^2) \propto (\sigma^2)^{-1}$

* Consider

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}} / \sqrt{n^*}}, \quad \text{where } n^* = \frac{n_1 n_2}{n_1 + n_2}$$

• Properties of BF

– Generally we need proper priors of $[\theta_1 | M_1]$ and $[\theta_2 | M_2]$

– For simple hypothesis, it reduces to the likelihood ratios

– Unlike some other model comparison tools, it works for non-nested models

→ R2 package for calculating BF: BridgeSampling

→ In order to calculate $BF_{12} = \frac{p(y|M_1)}{p(y|M_2)}$, you need to calculate

this integral: $p_i(y|M_i) = \int p_i(y|\theta_i, M_i) \pi(\theta_i|M_i) d\theta_i$

⇒ Bridge sampling gives good approximations of this

→ In Rstan, can compare two models (run both of them first),
+ then do `bridge_sampler(mod1)`, which gives you the marginal
probability of each model. Then can get BF by taking the ratio

`bridge_sampler :: bridge_sampler(mod2)`
`bridge_sampler :: bridge_sampler(mod1)`